Katla Satyanarayana

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Address: Next Generation Wireless Group, University of Southampton,

Objective

Looking for an opportunity to get involved in a role where I can be at the forefront of the cutting edge technologies/research and also have an opportunity to be able to present my ideas that can become global standard.

Education

University of Southampton, UK *Mar'16 -till date*

Thesis: **Intelligence-Aided Transceiver Design for Millimeter Wave Communications**

Interests: **Machine Learning**, Millimeter Wave Communications, Massive MIMO, Beamforming, Internet-of-Things

Supervisors: Lajos Hanzo, Mohammed El-Hajjar

Indian Institute of Technology Madras, Chennai *Aug'10 - July'14*

Bachelor of Technology in Electrical Engineering

Thesis: Interference Cancellation and Block Modulation Grade: 8/10

Modules: Multicarrier Communications, Information Theory, Adaptive Signal Processing, Digital Communications, Communication Systems, Communication Techniques, Probability and Stochastic process, Networks and Systems, Analog and Digital Signal Processing, Digital Systems.

Research Experience

Intelligence-Aided Transceiver Design for mmWave Communications

Supervisors: Prof. Lajos Hanzo and Prof. Mohammed El-Hajjar, University of Southampton *Mar'16- Till Date*

- ◇ Proposed transceiver design for full duplex mmWave communications relying on **machine learning** techniques.
- ◇ Devised channel prediction algorithm for systems suffering from the channel aging phenomenon by invoking sophisticated deep learning aided neural networks, namely **radial basis neural network**.
- ◇ To circumvent the search complexity involved in the beam-alignment schemes, a logistic regression function, namely the **soft-max aided neural network** is proposed to efficiently align the beam-pairs.
- ◇ Furthermore, to reduce the pilot-overhead involved in the channel estimation, a semi-blind detection is proposed relying on the **continuous regression neural network**. The propose design reduces the overhead to 0.002%.
- ◇ Additionally, learning-aided link-adaptation for C-RAN systems is proposed, where the adaptation is carried out intelligently to decide on the number of remote radio heads that communicate with the user while also deciding on the mode of transmission.
- ◇ Co-supervised three Ph.D. students, where the work resulted in 1 IEEE peer-reviewed conference and 1 IET journal.

Spatial Modulation

Supervisor: Prof. Chandra R. Murthy, IISc Bangalore *Jul'14- Aug'15*

- ◇ Designed precoders which maximizes the lower bound on mutual information of spatial modulation for finite alphabets.
- ◇ Compared critically spatial modulation with transmit antenna selection in terms of mutual information, outage probability, symbol error rate.

Gigabit Passive Optical Networking (GPON),

Alcatel-Lucent India Pvt Ltd., Chennai *May-Jul'13*

- ◇ Worked on Optical Networking Terminal (ONT) Management Control Interface (OMCI) which is a protocol between the Optical Line Terminal (OLT) and Optical Network Terminal (ONT).
- ◇ Performed sanity test for various ONT software builds using Spiral Test Center (STC) traffic generator.
- ◇ Studied various member entities for ONT such as Voice over Internet Protocol (VoIP), Video on Demand (VoD), Plain Old Telephonic Service (POTS), IPTV, Ethernet.
- ◇ Studied extensively on Internet Group Management Protocol (IGMPv2, IGMPv3).
- ◇ Also, carried out an independent study on Future Proof Fiber Access Passive Networks: Wavelength Division Multiplexing Passive Optical Network.

Implementation of Amplitude Modulation on Software Defined Radio,

Supervisors: Suneel Madhekar, Scientist 'F'

Defence Research and Development Organization, Ministry of Defence, Govt. of India *May-July'12*

- ◇ Implemented Amplitude Modulation on a Software Defined Radio.
- ◇ Interfaced Digital Signal Processor with Virtex-4 FPGA and with RF section of SDR
- ◇ Designed FIR filters for Virtex-4 FPGA
- ◇ Observed organization operations employing spread spectrum.

Education Projects

Interference Cancellation and Block Modulation
Guide: Prof. K. Giridhar, IIT-Madras *Jul'13-Jun'14*

- ◇ Simulated both the coded and uncoded BER using the designed precoders for SISO.
- ◇ LTE release 10 pilot structure is used for 10 physical resource blocks.
- ◇ Compared the ICBM scheme (3 users) of symbol rate 1/2 (two symbols for four resources) with reuse factor 1 matching the information rates.
- ◇ Turbo codes are used for coding gain, and aswell as to match the information rates of both the schemes.
- ◇ Also worked on maximizing the SINR based on Interferene Alignement Schemes, my undergraduate thesis can be found *here*.

Implementation of near far CDMA on Software Defined Radio
Guide: Prof. K. Giridhar, IIT-Madras *Feb-April'13*

- ◇ Three SDRs were used one as receiver and the other two as transmitters
- ◇ Signals of different power were sent by the transmitters with the SDR distance from the receiver remains the same.
- ◇ Orthogonal Codes were used for spreading the data (signal).
- ◇ Performed auto-correlation on received signals to retrieve the signal and analyzed it on Matlab.

Characterization of noise received by Software Defined Radio,
Guide: Prof. K. Giridhar, IIT-Madras *December 2012*

- ◇ Collected the samples of the noise from the antenna
- ◇ Performed auto-correlation on noise samples
- ◇ Power spectral density (PSD) of noise was calculated by performing Fourier transform of the autocorrelation function on noise samples
- ◇ Observed the noise was colored and its autocorrelated function was a **sinc** with second lobe approximately 13dB below the first lobe.

Microcontroller Network test,
IIT-Madras Satellite team *Fall 2011*

- ◇ Tested the buses and peripherals of Texas Instruments MSP430 microcontrollers, for the Command and Data Management Sub-system (CDMS) of IITMSAT
- ◇ Worked with a team on the network topology for the microcontrollers using I2C bus.
- ◇ Evaluated the performance of individual protocols with various topologies like multi slave and single master et al.

Mentor at Young Engineers,
R.K.A Technologies Pvt Ltd., Hyderabad *May-July'11*

- ◇ Worked in a team to simulate and program the autonomous vacuum cleaner robot
- ◇ Successfully trained robot to pick up the dust and dump it into a box attached on the chassis
- ◇ Responsibilities include leading the class of 120 students and conducting robotic sessions.

Awards and Honors

- IRACON-WaveComBE, grant for attending workshop in Dublin, Ireland, Jan'19
- IRACON, ISP-IOT, workshop grant,Lyon, France, Nov'18
- IRACON-Met5G, grant for attending training school in Surrey, UK, April'18
- IRACON-COST, grant for attending winter school in Lison, Portugal, March'18
- Runner up in the Southampton University's 3-minute thesis (3MT) competition.
- **InterDigital Scholarship**, University of Southampton *2016 - 2019*
- **Merit Scholarship holder**, Indian Institute of Technology-Madras *2010 - 2014*

Workshops and Conferences Attended

- ◇ Information Theory and Signal Processing for Internet of Things, IRACON, Lyon, France, 2018
- ◇ Defining and Realising SINR in Future Mobile Networks, IRACON/Met5G, University of Surrey, UK, 2018
- ◇ Beyond 5G Networks Operating in the Millimeter Wave bands enabled by the Joint Analog-Digital Signal Processing, COST-IRACON, Lisbon, 2018
- ◇ Personal, Indoor and Mobile Radio Communications, Bologna, 2018
- ◇ Vehicular Technology Conference, Toronto, 2017
- ◇ International Conference on Communication, Paris, 2017
- ◇ European Signal Processing Conference, Budapest, 2016

Professional Activities and Presentations

- ◇ Reviewer for IEEE Transactions on Wireless Communications
- ◇ Reviewer for IEEE Transactions on Vehicular Technology
- ◇ Reviewer for IET Communcations
- ◇ Co-supervised two senior undergraduates at University of Southampton
- ◇ Member of Institute of Electrical and Electronic Engineers (IEEE)
- ◇ Member of International Association of Engineers (IAENG)
- ◇ Member of IAENG society of Electrical Engineering, Wireless Networks, and Imaging Engineering
- ◇ Demonstrator and Mentor for undergraduate laboratories
- ◇ Presented a model for waterless washing machine and magnetic levitation at LG Industry Defined Problems, Indian Institute of Technology Madras.

Invited Talks

- ◇ Invited speaker at Winchester Cafe Scientifique, UK
- ◇ Invited speaker at Cafe Scientifique, Southampton, UK
- ◇ Invited speaker at Basingstoke Cafe Scientific, UK

Skills

Programming: C++, C, Python, Matlab, Simulink, VHDL/Verilog, HTML, LaTeX, Assembly (ARM/8051)

Software tools:Adobe Photoshop, Autocad, Lt-spice, Eagle, Code Composer Studio, Xilinx (ISE)

Operating Systems: Windows, Linux (Ubuntu, Fedora)

Languages: Hindi, English, Telugu, German

Patents

1. Ping-Heng Kuo , Alain Mourad, **Katla Satyanarayana**, Mohammed El-Hajjar, *Adaptive Digital Precoder Codebook Configurations for mmWave Communication based on Hybrid Beamforming*, patent filed, 2018.
2. Ping-Heng Kuo , Alain Mourad, **Katla Satyanarayana**, Mohammed El-Hajjar, *Method of Non-Orthogonal Uplink Multiplexing of Heterogeneous Information Types*, patent filed, 2019.

Publications (Accepted & Preprints)

1. **K. Satyanarayana**, Mohammed El-Hajjar, Alain Mourad, Lajos Hanzo, **Multi-User Hybrid Beamforming Relying on Learning-Aided Link-Adaptation for mmWave Systems**in IEEE Access Feb'19.
2. **K. Satyanarayana**, Mohammed El-Hajjar, Alain Mourad, Lajos Hanzo, **Deep Learning Aided Fingerprint Based Beam Alignment for mmWave Vehicular Communication**, submitted to TVT.
3. **K. Satyanarayana**, Mohammed El-Hajjar, Ping-Heng Kuo, Alain Mourad, Lajos Hanzo, **Multi-User Full Duplex Transceiver Design for mmWave using Learning-Aided Channel Prediction**, in IEEE Access May'19.
4. Yichuan Li, **K. Satyanarayana**, Mohammed El-Hajjar, Lajos Hanzo, **Analogue Radio Over Fiber Aided MIMO Design for the Learning Assisted Adaptive C-RAN Downlink** accepted IEEE Access, Jan'19.
5. **K. Satyanarayana**, Mohammed El-Hajjar, Alain Mourad, Lajos Hanzo, **Deep Learning Aided Semi-Blind Detection for Index Modulation mmWave Systems**, under preparation.

6. **K. Satyanarayana**, Mohammed El-Hajjar, Ping-Heng Kuo, Alain Mourad, Lajos Hanzo, *Hybrid Beamforming Design for Full-Duplex Millimeter Wave Communication* accepted in IEEE Transactions on Vehicular Technology, Nov'18.
7. **K. Satyanarayana**, Mohammed El-Hajjar, Ping-Heng Kuo, Alain Mourad, Lajos Hanzo, *Adaptive Transceiver Design for C-RAN in mmWave Communications* in IEEE Access, vol. 6, Nov. 2017.
8. **K. Satyanarayana**, Mohammed El-Hajjar, Ping-Heng Kuo, Alain Mourad, Lajos Hanzo, *Dual-Function Hybrid Beamforming and Transmit Diversity Aided Millimeter Wave Architectures* in IEEE Transactions on Vehicular Technology, vol. 67, no. 3, Aug. 2017.
9. **K. Satyanarayana**, Teodor Ivanescu, Mohammed El-Hajjar, Ping-Heng Kuo, Alain Mourad, Lajos Hanzo, *Hybrid Beamforming Design for Dual-Polarized Millimeter Wave MIMO Systems*, in IET Electronics Letter.
10. **K. Satyanarayana**, Mohammed El-Hajjar, Ping-Heng Kuo , Alain Mourad, Lajos Hanzo, *MBER Transmit Precoding for the Rank-Deficient MIMO-Aided Internet of Things*, in Proc. IEEE Personal, Indoor and Mobile Radio Communications (PIMRC'18).
11. **K. Satyanarayana**, Mohammed El-Hajjar, Ping-Heng Kuo , Alain Mourad, Lajos Hanzo, *Millimeter Wave Hybrid Beamforming with DFT-MUB Aided Precoder Codebook Design*, in Proc. IEEE Vehicular Technology Conference (VTC'17).
12. Ibrahim A. Hemadeh, **K. Satyanarayana**, Mohammed El-Hajjar, and Lajos Hanzo *Next-Generation Millimeter-Wave Communications: Propagation Characteristics, Channel Models, Antenna Constructions and Link-Budget* in IEEE Communications Surveys and Tutorials, vol. 20, no. 2, Dec. 2017.
13. Denisa Prisiceanu, **K. Satyanarayana**, Mohammed El-Hajjar, Ping-Heng Kuo , Alain Mourad, Lajos Hanzo, *Lattice Reduction-Aided Detection for Millimeter Wave Hybrid Beamforming with Mutual Coupling*, in Proc. IEEE Personal, Indoor and Mobile Radio Communications (PIMRC'18).
14. Yichuan Li, Salman Ghafoor, **K. Satyanarayana**, Mohammed El-Hajjar, Lajos Hanzo, *Analogue Wireless Beamforming Exploiting the Fiber-Nonlinearity of Radio Over Fiber Based C-RANs* accepted in IEEE Transactions on Vehicular Technology, Jan'19.
15. **K. Satyanarayana**, Suneel Madhekar, Ravi Ranjan. "*Implementation of Amplitude Modulation on Software Defined Radio,*", in *Proceedings of International Conference on Advanced Computing & Communication Technologies*. Panipat. 2012.
16. **K. Satyanarayana**, Abhinov Balagoni. "*Recent Trends in Future-Proof Fiber Access Passive Networks: GPON and WDM PON" Fourth International Conference on Recent Trends in Information Technology,India.*. Chennai. 2014. available at *arXiv 1308.5356*

References

Prof. Lajos Hanzo (Email: lh@ecs.soton.ac.uk)
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